

Justify your answers and show your work.

1. Let $\int_2^4 f(x) dx = 3$ and $\int_4^6 f(x) dx = -2$. Find $\int_6^2 f(x) dx$.

2. Find the indefinite integral of $f(x) = \frac{x^2 + 1}{x^2}$.

3. Evaluate $\int_0^{\pi/2} (\sin \theta + \cos \theta) d\theta$.

4. (**HW** 5.4: 23) What is the average distance between the parabola $y = 200x - 10x^2$ and the x -axis on the interval $[0, 20]$? (Your answer can be a decimal approximated to one-hundredths.)

5. Use the substitution rule to evaluate $\int x(x^2 + 10)^{30} dx$.